

When uncertain inputs become hard decisions

Planning and prioritisation often rely on fixed thresholds and rankings. Habnetic estimates whether those decisions remain stable when uncertainty is propagated through the decision rule.

WHY DECISION STABILITY MATTERS

Risk estimates are uncertain, but planning decisions are often discrete: prioritise / do not prioritise.

Habnetic propagates posterior uncertainty through that decision rule. Most assets remain clearly classified; uncertainty is concentrated around a small decision boundary.

In Rotterdam, more than 99% of buildings remain outside that unstable boundary.

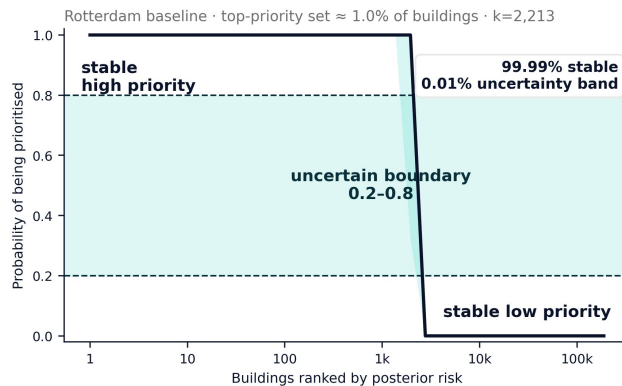
Posterior probability of top-k membership – ROTTERDAM



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DECISION STABILITY STRUCTURE



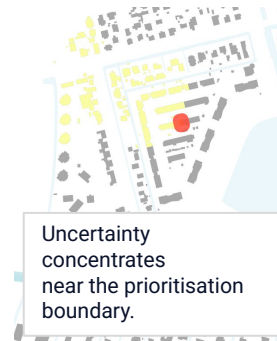
ZOOMED EXAMPLE ROTTERDAM

A. Consistently high-priority area



Most assets have very low or very high probability → stable decisions.

B. Boundary area (mixed uncertainty)



Uncertainty concentrates near the prioritisation boundary.

CROSS CITY TRANSFER

Same model, different cities



Hamburg



Donostia / SS

MODEL FRAMEWORK



WHAT THIS ENABLES



Identify unstable decisions



Target review at boundary cases



Distinguish robust from fragile priorities

DECISION IMPLICATION



Do not review uncertainty everywhere. Review where it can change the decision.



Paper & data
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